形式态射 VERSALITY 的统一光滑邻域

Editorial reading note. The printed reversal A_n to B_n in the final sentence is retained. The preceding argument concerns B_n to A_n.

Counting scope. The primary problem is Tag 0GI9, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Difficulty H2: 本题从一点 versality 推出存在一个对所有形式厚化层同时有效的光滑开邻域。各层分别光滑只给随 n 变化的开集；证明把所有分次层装入 Rees 代数映射，并利用核余核有限性在一次局部化后同时成为同构。这个统一化步骤及 étale 代数化归约仍由本证明完成，判 H2。.

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1. Axioms for functors

Let S be a scheme. Let $F : (Sch/S)^{opp}_{fppf} \to Sets$ be a functor. Denote $\mathcal{X} = \mathcal{S}_F$ the category fibred in sets associated to F, see Algebraic Stacks, Section section split (Tag 04SU). In this section we provide a translation between the material above as it applies to $\mathcal{X}$, to statements about F.

Let S be a locally Noetherian scheme. Let $F : (Sch/S)^{opp}_{fppf} \to Sets$ be a functor. Let k be a field of finite type over S. Let $x_0 \in F(\mathrm{Spec}(k))$. The associated predeformation category (equation predeformation category (Tag 07T4)) corresponds to the functor

$$F_{k,x_0} : \mathcal{C}_\Lambda \longrightarrow Sets, \quad A \longmapsto \{x \in F(\mathrm{Spec}(A)) \mid x|_{\mathrm{Spec}(k)} = x_0\}.$$

Recall that we do not distinguish between categories cofibred in sets over $\mathcal{C}_\Lambda$ and functor $\mathcal{C}_\Lambda \to Sets$, see Formal Deformation Theory, Remarks remarks cofibered groupoids (Tag 06GK) (item convention cofibered sets (Tag 06GP)). Given a transformation of functors $a : F \to G$, setting $y_0 = a(x_0)$ we obtain a morphism

$$F_{k,x_0} \longrightarrow G_{k,y_0}$$

see (equation functoriality (Tag 07WJ)). Lemma lemma formally smooth on deformation categories (Tag 07WK) tells us that if $a : F \to G$ is formally smooth (in the sense of More on Morphisms of Spaces, Definition 1.3), then $F_{k,x_0} \longrightarrow G_{k,y_0}$ is smooth as in Formal Deformation Theory, Remark remark compare smooth schlessinger (Tag 06HJ).

Lemma lemma pushout (Tag 07WN) says that if $Y' = Y \amalg_X X'$ in the category of schemes over S where $X \to X'$ is a thickening and $X \to Y$ is affine, then the map

$$F(Y \amalg_X X') \to F(Y) \times_{F(X)} F(X')$$

is a bijection, provided that F is an algebraic space. We say a general functor F satisfies the *Rim-Schlessinger condition* or we say F *satisfies (RS)* if given any pushout $Y' = Y \amalg_X X'$ where Y, X, X' are spectra of Artinian local rings of finite type over S, then

$$F(Y \amalg_X X') \to F(Y) \times_{F(X)} F(X')$$

is a bijection. Thus every algebraic space satisfies (RS).

Lemma lemma deformation category (Tag 07WU) says that given a functor F which satisfies (RS), then all F_{k,x_0} are deformation functors as in Formal Deformation Theory, Definition 1.4, i.e., they satisfy (RS) as in Formal Deformation Theory, Remark remark compare schlessinger H4 (Tag 06J6). In particular the tangent space

$$TF_{k,x_0} = \{x \in F(\mathrm{Spec}(k[\epsilon])) \mid x|_{\mathrm{Spec}(k)} = x_0\}$$

has the structure of a k-vector space by Formal Deformation Theory, Lemma lemma tangent space vector space (Tag 06IH).

Lemma lemma finite dimension (Tag 07X1) says that an algebraic space F locally of finite type over S gives rise to deformation functors F_{k,x_0} with finite dimensional tangent spaces TF_{k,x_0}.

A *formal object*[1] $\xi = (R, \xi_n)$ of F consists of a Noetherian complete local S-algebra R whose residue field is of finite type over S, together with elements $\xi_n \in F(\mathrm{Spec}(R/\mathfrak{m}^n))$ such that $\xi_{n+1}|_{\mathrm{Spec}(R/\mathfrak{m}^n)} = \xi_n$. A formal object ξ defines a formal object ξ of $F_{R/\mathfrak{m},\xi_1}$. We say ξ is *versal* if and only if it is versal in the sense of Formal Deformation Theory, Definition 1.5. A formal object $\xi = (R, \xi_n)$ is called *effective* if there exists an $x \in F(\mathrm{Spec}(R))$ such that $\xi_n = x|_{\mathrm{Spec}(R/\mathfrak{m}^n)}$ for all $n \geq 1$. Lemma lemma effective (Tag 07X8) says that if F is an algebraic space, then every formal object is effective.

Let U be a scheme locally of finite type over S and let $x \in F(U)$. Let $u_0 \in U$ be a finite type point. We say that x is versal at u_0 if and only if $\xi = (\mathcal{O}^\wedge_{U,u_0}, x|_{\mathrm{Spec}(\mathcal{O}_{U,u_0}/\mathfrak{m}^n_{u_0})})$ is a versal formal object in the sense described above.

Let S be a locally Noetherian scheme. Let $F : (Sch/S)^{opp}_{fppf} \to Sets$ be a functor. Here are the axioms we will consider on F.

- [-1] a set theoretic condition[2] to be ignored by readers who are not interested in set theoretical issues,
- [0] F is a sheaf for the étale topology,
- [1] F is limit preserving,
- [2] F satisfies the Rim-Schlessinger condition (RS),
- [3] every tangent space TF_{k,x_0} is finite dimensional,
- [4] every formal object is effective,
- [5] F satisfies openness of versality.

Here *limit preserving* is the notion defined in Limits of Spaces, Definition 1.6 and *openness of versality* means the following: Given a scheme U locally of finite type over S, given $x \in F(U)$, and given a finite type point $u_0 \in U$ such that x is versal at u_0, then there exists an open neighbourhood $u_0 \in U' \subset U$ such that x is versal at every finite type point of U'.

Lemma 1.1. *Let S, $\mathcal{X}$, U, x, u_0 be as in Definition 1.7. Assume*

1. *$\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces,*
2. *Δ is locally of finite type (for example if $\mathcal{X}$ is limit preserving), and*
3. *$\mathcal{X}$ has (RS).*

[1] This is what Artin calls a formal deformation.

[2] The condition is the following: the supremum of all the cardinalities $|F(\mathrm{Spec}(k))|$ where k runs over the finite type fields over S is $\leq$ than the size of some object of $(Sch/S)_{fppf}$.

Let V be a scheme locally of finite type over S and let y be an object of $\mathcal{X}$ over V. Form the 2-fibre product

$$\begin{array}{ccc} \mathcal{Z} & \longrightarrow & (Sch/U)_{fppf} \\ \downarrow & & \downarrow{\scriptstyle x} \\ (Sch/V)_{fppf} & \xrightarrow{y} & \mathcal{X} \end{array}$$

Let Z be the algebraic space representing $\mathcal{Z}$ and let $z_0 \in |Z|$ be a finite type point lying over u_0. If x is versal at u_0, then the morphism $Z \to V$ is smooth at z_0.

Lemma 1.2. *Let $T \subset X$ be a closed subset of a Noetherian affine scheme X. Let W be a Noetherian affine formal algebraic space. Let $g : W \to X_{/T}$ be a rig-étale morphism. Then there exists an affine scheme X' and a finite type morphism $f : X' \to X$ étale over $X \setminus T$ such that there is an isomorphism $X'_{/f^{-1}T} \cong W$ compatible with $f_{/T}$ and g. Moreover, if $W \to X_{/T}$ is étale, then $X' \to X$ is étale.*

证明. The existence of X' is a restatement of Lemma lemma approximate by etale over complement (Tag 0AKG). The final statement follows from More on Morphisms, Lemma lemma check smoothness on infinitesimal nbhds (Tag 0A43). □

Definition 1.3. Let S be a scheme. Let $a : F \to G$ be a transformation of functors $F, G : (Sch/S)^{opp}_{fppf} \to Sets$. Consider commutative solid diagrams of the form

$$\begin{array}{ccc} F & \longleftarrow & T \\ {\scriptstyle a}\downarrow & \nwarrow & \downarrow{\scriptstyle i} \\ G & \longleftarrow & T' \end{array}$$

where T and T' are affine schemes and i is a closed immersion defined by an ideal of square zero.

(1) We say a is *formally smooth* if given any solid diagram as above there exists a dotted arrow making the diagram commute[3].
(2) We say a is *formally étale* if given any solid diagram as above there exists exactly one dotted arrow making the diagram commute.
(3) We say a is *formally unramified* if given any solid diagram as above there exists at most one dotted arrow making the diagram commute.

Definition 1.4. A *deformation category* is a predeformation category $\mathcal{F}$ satisfying (RS). A morphism of deformation categories is a morphism of categories over $\mathcal{C}_\Lambda$.

Definition 1.5. Let $\mathcal{F}$ be a category cofibered in groupoids. Let ξ be a formal object of $\mathcal{F}$ lying over $R \in \mathrm{Ob}(\widehat{\mathcal{C}}_\Lambda)$. We say ξ is *versal* if the corresponding morphism $\underline{\xi} : \underline{R}|_{\mathcal{C}_\Lambda} \to \mathcal{F}$ of Remark remark formal objects yoneda (Tag 06HC) is smooth.

Definition 1.6. Let S be a scheme.

(1) A functor $F : (Sch/S)^{opp}_{fppf} \to Sets$ is said to be *limit preserving* or *locally of finite presentation* if for every affine scheme T over S which is a limit $T = \lim T_i$ of a directed inverse system of affine schemes T_i over S, we have

$$F(T) = \operatorname{colim} F(T_i).$$

We sometimes say that F is *locally of finite presentation over S*.

[3]This is just one possible definition that one can make here. Another slightly weaker condition would be to require that the dotted arrow exists fppf locally on T'. This weaker notion has in some sense better formal properties.

(2) Let $F, G : (Sch/S)^{opp}_{fppf} \to Sets$. A transformation of functors $a : F \to G$ is *limit preserving* or *locally of finite presentation* if for every scheme T over S and every $y \in G(T)$ the functor

$$F_y : (Sch/T)^{opp}_{fppf} \longrightarrow Sets, \quad T'/T \longmapsto \{x \in F(T') \mid a(x) = y|_{T'}\}$$

is locally of finite presentation over T[4]. We sometimes say that F is *relatively limit preserving* over G.

Definition 1.7. Let S be a locally Noetherian scheme. Let $\mathcal{X}$ be fibred in groupoids over $(Sch/S)_{fppf}$. Let U be a scheme locally of finite type over S. Let x be an object of $\mathcal{X}$ lying over U. Let u_0 be a finite type point of U. We say x is *versal* at u_0 if the morphism $\hat{x}$ (equation hat x (Tag 07XE)) is smooth, see Formal Deformation Theory, Definition 1.9.

Definition 1.8. Let S be a scheme. Let $a : F \to G$ be a map of presheaves on $(Sch/S)_{fppf}$ which is representable by algebraic spaces. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces which

(1) is preserved under any base change, and
(2) is fppf local on the base, see Descent on Spaces, Definition 1.10.

In this case we say that a has *property* $\mathcal{P}$ if for every scheme U and $\xi : U \to G$ the resulting morphism of algebraic spaces $U \times_G F \to U$ has property $\mathcal{P}$.

Definition 1.9. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of categories cofibered in groupoids over $\mathcal{C}_\Lambda$. We say φ is *smooth* if it satisfies the following condition: Let $B \to A$ be a surjective ring map in $\mathcal{C}_\Lambda$. Let $y \in \mathrm{Ob}(\mathcal{G}(B)), x \in \mathrm{Ob}(\mathcal{F}(A))$, and $y \to \varphi(x)$ be a morphism lying over $B \to A$. Then there exists $x' \in \mathrm{Ob}(\mathcal{F}(B))$, a morphism $x' \to x$ lying over $B \to A$, and a morphism $\varphi(x') \to y$ lying over $\mathrm{id} : B \to B$, such that the diagram

$$\begin{array}{ccc} \varphi(x') & \longrightarrow & y \\ & \searrow & \downarrow \\ & & \varphi(x) \end{array}$$

commutes.

Definition 1.10. Let S be a scheme. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces over S. Let $\tau \in \{fpqc, fppf, syntomic, smooth, étale\}$. We say $\mathcal{P}$ is τ *local on the base*, or τ *local on the target*, or *local on the base for the* τ*-topology* if for any τ-covering $\{Y_i \to Y\}_{i \in I}$ of algebraic spaces and any morphism of algebraic spaces $f : X \to Y$ we have

$$f \text{ has } \mathcal{P} \Leftrightarrow \text{each } Y_i \times_Y X \to Y_i \text{ has } \mathcal{P}.$$

Primary problem: Tag 0GI9

Lemma 1.11. *Let S be a locally Noetherian scheme. Let V be a scheme locally of finite type over S. Let $Z \subset V$ be closed. Let W be a locally Noetherian formal algebraic space over S such that W_{red} is locally of finite type over S. Let $g : V_{/Z} \to W$ be an adic morphism of formal algebraic spaces over S. Let $v \in V$ be a closed point such that g is versal at v (as in Section 1). Then after replacing V by an open neighbourhood of v the morphism g is smooth (see proof).*

[4]The characterization (2) in Lemma lemma characterize relative limit preserving (Tag 06BC) may be easier to parse.

证明. Since g is adic it is representable by algebraic spaces (Formal Spaces, Section section adic (Tag 0AQ2)). Thus by saying g is smooth we mean that g should be smooth in the sense of Bootstrap, Definition 1.8.

Write $W = \operatorname{colim} W_n$ as in Formal Spaces, Lemma lemma structure locally noetherian (Tag 0GHM). Set $V_n = V_{/Z} \times_{\hat{x},W} W_n$. Then V_n is a closed subscheme with underlying set Z. Smoothness of $V \to W$ is equivalent to the smoothness of all the morphisms $V_n \to W_n$ (this holds because any morphism $T \to W$ with T a quasi-compact scheme factors through W_n for some n). We know that the morphism $V_n \to W_n$ is smooth at v by Lemma 1.1[5]. Of course this means that given any n we can shrink V such that $V_n \to W_n$ is smooth. The problem is to find an open which works for all n at the same time.

The question is local on V, hence we may assume $S = \operatorname{Spec}(R)$ and $V = \operatorname{Spec}(A)$ are affine.

In this paragraph we reduce to the case where W is an affine formal algebraic space. Choose an affine formal scheme W' and an étale morphism $W' \to W$ such that the image of v in W_{red} is in the image of $W'_{red} \to W_{red}$. Then $V_{/Z} \times_{g,W} W' \to V_{/Z}$ is an adic étale morphism of formal algebraic spaces over S and $V_{/Z} \times_{g,W} W'$ is an affine formal algebraic space. By Algebraization of Formal Spaces, Lemma 1.2 there exists an étale morphism $\varphi : V' \to V$ of affine schemes such that the completion of V' along $Z' = \varphi^{-1}(Z)$ is isomorphic to $V_{/Z} \times_{g,W} W'$ over $V_{/Z}$. Observe that v is the image of some $v' \in V'$. Since smoothness is preserved under base change we see that $V'_n \to W'_n$ is smooth for all n. In the next paragraph we show that after replacing V' by an open neighbourhood of v' the morphisms $V'_n \to W'_n$ are smooth for all n. Then, after we replace V by the open image of $V' \to V$, we obtain that $V_n \to W_n$ is smooth by étale descent of smoothness. Some details omitted.

Assume $S = \operatorname{Spec}(R)$, $V = \operatorname{Spec}(A)$, $Z = V(I)$, and $W = \operatorname{Spf}(B)$. Let v correspond to the maximal ideal $I \subset \mathfrak{m} \subset A$. We are given an adic continuous R-algebra homomorphism

$$B \longrightarrow A^\wedge$$

Let $\mathfrak{b} \subset B$ be the ideal of topologically nilpotent elements (this is the maximal ideal of definition of the Noetherian adic topological ring B). Observe that $\mathfrak{b}A^\wedge$ and $IA^\wedge$ are both ideals of definition of the Noetherian adic ring $A^\wedge$. Also, $\mathfrak{m}A^\wedge$ is a maximal ideal of $A^\wedge$ containing both $\mathfrak{b}A^\wedge$ and $IA^\wedge$. We are given that

$$B_n = B/\mathfrak{b}^n \to A^\wedge/\mathfrak{b}^n A^\wedge = A_n$$

is smooth at $\mathfrak{m}$ for all n. By the discussion above we may and do assume that $B_1 \to A_1$ is a smooth ring map. Denote $\mathfrak{m}_1 \subset A_1$ the maximal ideal corresponding to $\mathfrak{m}$. Since smoothness implies flatness, we see that: for all $n \geq 1$ the map

$$\mathfrak{b}^n/\mathfrak{b}^{n+1} \otimes_{B_1} (A_1)_{\mathfrak{m}_1} \longrightarrow \left(\mathfrak{b}^n A^\wedge/\mathfrak{b}^{n+1}A^\wedge\right)_{\mathfrak{m}_1}$$

is an isomorphism (see Algebra, Lemma lemma what does it mean again (Tag 0AS8)). Consider the Rees algebra

$$B' = \bigoplus\nolimits_{n \geq 0} \mathfrak{b}^n/\mathfrak{b}^{n+1}$$

[5]The lemma applies since the diagonal of W is representable by algebraic spaces and locally of finite type, see Formal Spaces, Lemma lemma diagonal morphism formal algebraic spaces (Tag 0AN2) and we have seen that W has (RS) in the proof of Lemma lemma rs (Tag 0GI6).

which is a finite type graded algebra over the Noetherian ring B_1 and the Rees algebra

$$A' = \bigoplus\nolimits_{n \geq 0} \mathfrak{b}^n A^\wedge / \mathfrak{b}^{n+1} A^\wedge$$

which is a a finite type graded algebra over the Noetherian ring A_1. Consider the homomorphism of graded A_1-algebras

$$\Psi : B' \otimes_{B_1} A_1 \longrightarrow A'$$

By the above this map is an isomorphism after localizing at the maximal ideal $\mathfrak{m}_1$ of A_1. Hence $\mathrm{Ker}(\Psi)$, resp. $\mathrm{Coker}(\Psi)$ is a finite module over $B' \otimes_{B_1} A_1$, resp. A' whose localization at $\mathfrak{m}_1$ is zero. It follows that after replacing A_1 (and correspondingly A) by a principal localization we may assume Ψ is an isomorphism. (This is the key step of the proof.) Then working backwards we see that $B_n \to A_n$ is flat, see Algebra, Lemma lemma what does it mean again (Tag 0AS8). Hence $A_n \to B_n$ is smooth (as a flat ring map with smooth fibres, see Algebra, Lemma lemma flat fibre smooth (Tag 00TF)) and the proof is complete. □